

Module 4: Dashboard Planning and Prototyping

Unified Military Analytics and Comparison Dashboard

Author: Yogeshwar Vadla

Internship: Infosys Springboard

Domain: Data Visualization

Methodology

Module 4 focused on transforming the analytics-ready dataset (from Module 3) into a structured, user-centric dashboard design through systematic planning and prototyping.

Step 1: Storyboarding & Wireframe Design

Before building the dashboard in Power BI, draft layout sketches (wireframes) were created for the following pages:

➤ Quick Stats

- High-level global KPIs
- Budget-to-GDP indicator
- Defense budget gauge
- Defense budget per capita
- Military manpower summary
- Regional asset distribution
- Global power ranking overview
- Population vs manpower comparison
- World map visualization

➤ Nation Overview

- Country-specific KPI breakdown
- Detailed manpower, land, air, naval power visuals
- Economic and infrastructure metrics
- Alliance and regional classification

➤ ♦ Compare Powers

- Side-by-side country comparison
- KPI comparison charts
- Budget and manpower contrast
- Rank gap analysis

➤ **Coalition Builder**

- Multi-country selection logic
- Combined military strength calculation
- Aggregated defense budget and manpower
- Alliance simulation capability

These wireframes defined visual hierarchy, KPI placement, user flow, and navigation structure before implementation.

Step 2: Defining Filters, Buttons & Navigation Logic

The following interactive elements were planned and implemented:

Filters

- Country search box
- Country selection slicer
- NATO membership filtering

Buttons

- Page navigation arrows
- KPI drill-through logic
- Reset/clear selection functionality

Navigation Logic

- Centralized filter behaviour across pages
- Consistent theme and KPI positioning
- Smooth drill-through between summary and detailed views

This ensured the dashboard is user-driven and analytically flexible.

Step 3: Prototype Dashboard Development (Quick Stats Page)

Using sample analytical data, a working dashboard prototype was developed in Power BI.

Quick Stats Prototype Includes:

1. **Top KPI Cards**
 - Budget-to-GDP vs Manpower
 - Defense Budget Gauge
 - Defense Budget per Capita
2. **Military Manpower Summary**
 - Aggregated global military manpower value
3. **Regional Asset Distribution**
 - Pie chart of average assets per capita by region

4. **Global Defense Power Overview**
 - Ranked horizontal bar chart
5. **Population vs Manpower Comparison**
 - Donut chart representation
6. **Global Map Visualization**
 - Geographic distribution of countries
7. **Defense Budget vs Manpower Trend**
 - Dual-axis line comparison

The layout follows a **top-to-bottom analytical storytelling structure**:

- Start with summary KPIs
- Move to ranking insights
- End with geographic and trend-based visuals

Insights from Quick Stats Prototype

The Quick Stats dashboard provides immediate macro-level strategic understanding:

Global-Level Insights

- Large economies dominate global defense spending.
- Military manpower is heavily concentrated in high-population countries.
- Budget allocation relative to GDP differs significantly across nations.
- Certain countries show high military density despite smaller populations.
- Regional clusters demonstrate defense concentration patterns.

Strategic Analytical Insights

- Budget-to-GDP ratio helps identify countries prioritizing defense.
- Assets per capita normalizes raw military size.
- Rank-based visuals highlight global power positioning.
- NATO-based filtering enables alliance comparison.
- The global map reveals continental defense concentration trends.

Design & UX Insights

- KPI-first layout improves executive readability.
- Dark theme enhances contrast and strategic presentation.
- Grouped visuals reduce cognitive overload.
- Navigation arrows allow scalable multi-page architecture.
- Combining numeric KPIs with visuals improves storytelling depth.

Module 4 Outcome

Module 4 successfully bridges analytical data engineering and interactive visualization by:

- Converting KPIs into dashboard-ready components
- Designing scalable multi-page dashboard architecture
- Implementing navigation logic
- Building a prototype application using sample data

This module establishes the foundation for final deployment and advanced analytical storytelling.